Lab 4:
Sampling with R

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Sampling

- **Sampling** (การสุ่มตัวอย่าง) is the selection of a subset of a population (a sample) to estimate characteristics of the population
- **Examples:**
  - We sample 50 bottled water of various brands in stores, and check the quality of the water according to drinking water quality standard
  - We flip a coin 10 times and count number of heads

Sampling with R (1)

- **`sample(x, size, replace = FALSE, prob = NULL)`**
  - Select a random sample from the population provided
- Rolling a die:

```
A <- 1:6
sample(A, 1)
```

- Rolling 8 dice:

```
A <- 1:6
sample(A, 8, replace = T)
```

Sampling with R (2)

- Flipping a coin:

```
A <- c("H", "T")
sample(A, 1)
```

- Flipping a biased coin:

```
A <- c("H", "T")
P <- c(0.2, 0.8)
sample(A, 1, prob = P)
```

Counting Elements in a Vector

- `==` is the comparison operator:

```
A <- c(5, 8, 2, 0, 8, 0, 2, 3, 5, 8)
print(A == 8)
```

- Count using sum and `==`:

```
A <- c(5, 8, 2, 0, 8, 0, 2, 3, 5, 8)
x <- sum(A == 8)
print(x)
```

All Comparison Operators

- <, <=, >, >=, ==, !=